

EX-10 Content Creation Using Prompt Patterns

Name: R. Divyadharshini

Register Number: 212225230062

Subject: 19TD608 - Prompt Engineering

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Aim

To demonstrate how different prompt structures influence the quality, creativity, coherence, and organization of AI-generated content using various prompt engineering techniques.

Objective

- To create prompts for content generation.
- To generate reports, articles, and creative content using AI tools.
- To compare outputs generated using different prompt patterns.
- To analyze the effectiveness of prompt engineering techniques.

AI Tools Used

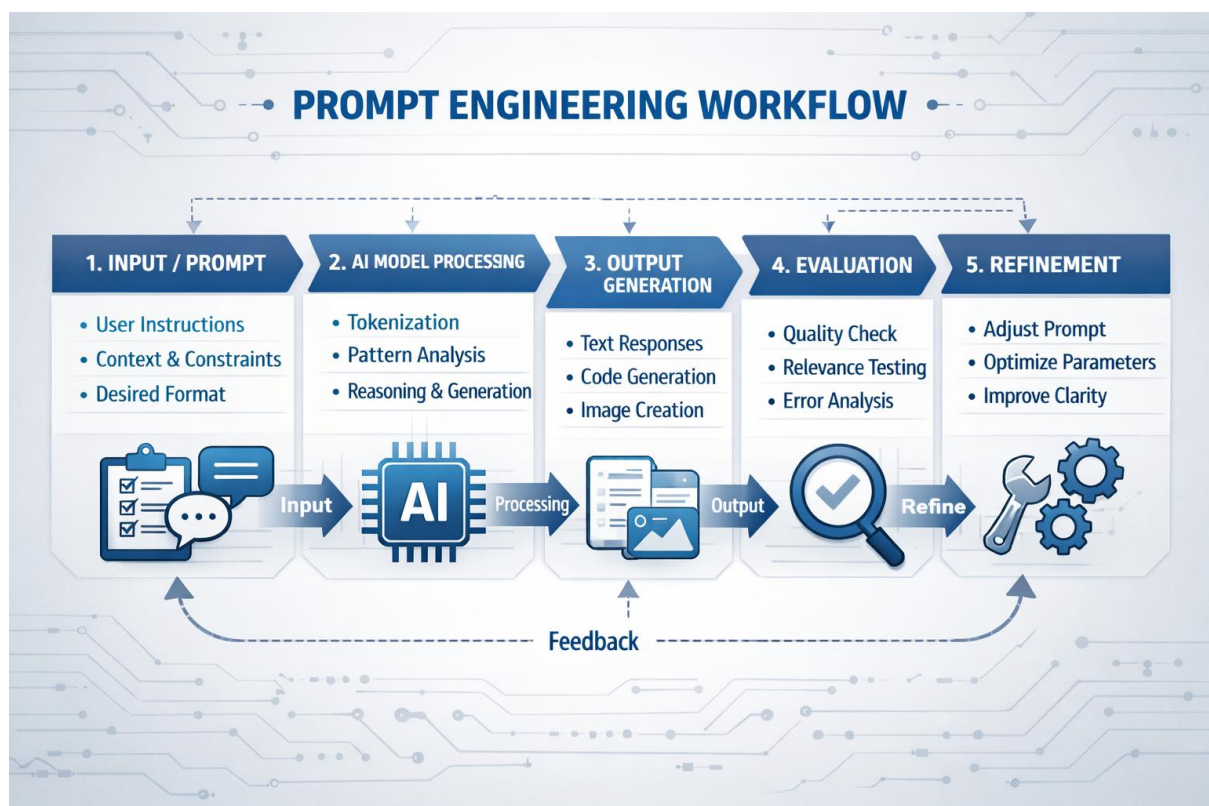
1. ChatGPT
2. Gemini

Prompting Techniques Used

1. Query Decomposition
2. Decision-Making Prompts
3. Semantic Filtering
4. Structured Prompting
5. Creative Prompting

Problem Statement

Create different types of content using various prompt engineering techniques and analyze how prompt structure affects the generated output.



Technique 1 – Query Decomposition

Initial Prompt

“Write an article about Artificial Intelligence.”

Improved Prompt Using Query Decomposition

“Explain Artificial Intelligence by dividing the topic into the following sections:

1. Definition
2. History
3. Applications
4. Advantages
5. Challenges
6. Future Scope”

ChatGPT Output

Artificial Intelligence

Definition

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines programmed to think and learn.

History

AI originated in the 1950s with the development of early computing systems and evolved rapidly with machine learning.

Applications

AI is used in healthcare, finance, education, transportation, and virtual assistants.

Advantages

AI improves efficiency, reduces human effort, and enhances decision-making.

Challenges

Privacy concerns, job displacement, and ethical issues remain major challenges.

Future Scope

AI is expected to revolutionize industries with automation and intelligent systems.

Gemini Output

Artificial Intelligence is a rapidly growing technology that enables machines to mimic human intelligence. It is widely used in many industries and provides automation and smart decision-making capabilities. However, ethical and privacy concerns still exist.

Analysis

Parameter	ChatGPT	Gemini
Structure	Highly organized	Moderately organized
Detail Level	Detailed explanation	Short explanation
Clarity	Excellent	Good
Readability	Very high	High
Overall Quality	Excellent	Good

Technique 2 – Decision-Making Prompt

Initial Prompt

“Suggest a career option.”

Improved Prompt Using Decision-Making Prompt

“I am a student interested in Artificial Intelligence, coding, and data analysis. Suggest the best career option with reasons, required skills, and future opportunities.”

ChatGPT Output

Recommended Career: Data Scientist

Reasons

- High demand in industries
- Strong alignment with AI and analytics
- Excellent salary opportunities

Required Skills

- Python Programming
- Machine Learning
- Data Visualization
- Statistics

Future Opportunities

Data Scientists are needed in healthcare, finance, business analytics, and automation industries.

Gemini Output

A good career option for you is Data Science because it combines coding, AI, and analytics. Learning Python, statistics, and machine learning will help you succeed

Analysis

Parameter	ChatGPT	Gemini
Personalization	High	Moderate
Explanation	Detailed	Brief
Decision Support	Excellent	Good
Content Quality	Excellent	Good

Technique 3 – Semantic Filtering

Initial Prompt

“Write about technology.”

Improved Prompt Using Semantic Filtering

“Write only about the positive impact of green technology in reducing environmental pollution. Avoid discussing disadvantages.”

ChatGPT Output

Green technology plays an important role in reducing environmental pollution. Renewable energy sources such as solar and wind power reduce carbon emissions. Electric vehicles help decrease air pollution, while recycling technologies minimize waste generation.

Gemini Output

Green technology helps create a cleaner environment through renewable energy, sustainable manufacturing, and eco-friendly transportation systems.

Analysis

Parameter	ChatGPT	Gemini
Topic Focus	Highly focused	Focused
Content Depth	Detailed	Moderate
Relevance	Excellent	Good
Output Quality	Excellent	Good

Technique 4 – Structured Prompting

Initial Prompt

“Write a report on cybersecurity.”

Improved Structured Prompt

“Create a report on cybersecurity with the following format:

- 1. Introduction
- 2. Importance
- 3. Common Cyber Threats
- 4. Prevention Methods
- 5. Conclusion”

ChatGPT Output

Cybersecurity Report

Introduction

Cybersecurity refers to the protection of computer systems and data from digital attacks.

Importance

It ensures data privacy, financial security, and safe online communication.

Common Cyber Threats

- Malware
- Phishing
- Ransomware
- Hacking

Prevention Methods

- Strong passwords
- Antivirus software
- Regular updates
- User awareness

Conclusion

Cybersecurity is essential for protecting digital information and maintaining online safety.

Gemini Output

Cybersecurity protects systems from threats such as hacking and malware. Using strong passwords and security software can help reduce risks.

Analysis

Parameter	ChatGPT	Gemini
Formatting	Excellent	Basic

Structure	Highly organized	Less organized
Detail	Detailed	Brief
Professional Quality	Excellent	Good

Technique 5 – Creative Prompting

Initial Prompt

“Write a story.”

Improved Creative Prompt

“Write a short science-fiction story about a student who invents an AI robot that can predict future events.”

ChatGPT Output

A college student named Arjun developed an AI robot called VisionX. The robot could analyze patterns and predict future events accurately. One day, VisionX warned about a major system failure in the city. Arjun informed the authorities, preventing a disaster and becoming a national hero.

Gemini Output

A student created a smart AI robot capable of predicting future situations. The invention became famous after it helped people avoid accidents and dangers.

Analysis

Parameter	ChatGPT	Gemini
Creativity	Excellent	Good

Story Development	Detailed	Short
Imagination	High	Moderate
Engagement	Excellent	Good

Overall Comparative Analysis

Parameter	ChatGPT	Gemini
Response Detail	More detailed	More concise
Creativity	Excellent	Good
Structured Output	Highly structured	Moderately structured
Clarity	Excellent	Good
User Interaction	Interactive	Less interactive
Professional Formatting	Excellent	Good
Overall Performance	Highly effective	Effective

Advantages of Prompt Engineering

- Improves response quality
- Enhances creativity and structure
- Produces focused and relevant outputs
- Helps generate professional content
- Saves time in content creation

Result

Thus, the experiment on “Content Creation Using Prompt Patterns” was successfully completed. Different prompt engineering techniques significantly improved the quality, structure, creativity, and relevance of AI-generated content.

References

1. OpenAI – ChatGPT
2. Google – Gemini
3. Prompt Engineering Guide
4. Artificial Intelligence Documentation